



Algerian Olympiad in Artificial Intelligence

AOAI Genesis

Drici Mohammed Hachem

AOAI Training Program

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1 Introduction

AOAI Genesis is the first stage of the second Algerian Olympiad in Artificial Intelligence (AOAI) season and is open to all participants who meet the eligibility requirements.

The purpose of this stage is to identify participants who can learn, adapt, reason, and work independently when faced with unfamiliar concepts and problems. The stage therefore focuses on mathematical reasoning, programming ability, adaptability, and the ability to acquire and apply new knowledge.

Participants will be assessed in two areas: **Mathematics** and **Programming**. The Mathematics test will cover concepts beyond the standard high school curriculum, including areas relevant to the mathematical foundations of Artificial Intelligence. The Programming test will use **Python** and will assess the ability to translate mathematical and algorithmic ideas into code and implement basic algorithms independently.

Participants should also be comfortable working with a modern development environment, including tools such as Visual Studio Code and Google Colab, and should be able to read and communicate technical ideas in English.

AOAI Genesis is **not intended to measure how much AI-specific knowledge a participant already possesses**. The tasks will provide the necessary context and concepts where appropriate. Instead, the stage is designed to assess whether participants have the ability and willingness to learn those concepts and use them to solve unfamiliar problems.

Approximately **50% of participants** will progress to the next stage of the Season 2 selection process. Participants who do not progress are still encouraged to continue studying the material provided for later stages, which will remain available to help them develop their skills and prepare for future selection opportunities.

2 Round Structure

AOAI Genesis consists of two separate tests: a **Mathematics Test** and a **Programming Test**.

Each test has a duration of **5 hours**. The Mathematics Test will take place first, followed by the Programming Test one week later.

Test	Date	Start Time	Duration
Mathematics Test	October 3, 2026	08:00	5 hours
Programming Test	October 10, 2026	08:00	5 hours

3 Test Format and Rules

Both AOAI Genesis tests will be conducted fully online.

For each test, a Google Form will be provided on the day of the test. The form will contain the submission instructions and any additional information required to submit the solutions correctly.

3.1 Rules and Restrictions

The tests must be completed individually. Participants are not permitted to receive assistance from other contestants or from any other person during the test.

The use of **any Artificial Intelligence tool is strictly forbidden**. This includes, but is not limited to:

- Large Language Models (LLMs), whether accessed through commercial services or run locally;
- AI-powered assistants integrated into text editors or development environments;
- AI-powered code completion or code generation tools;
- Any other tool that uses Artificial Intelligence to generate, explain, modify, or assist with solutions.

Participants must complete the tests using their own knowledge, reasoning, and work.

3.2 Allowed Resources

The resources that may be accessed during each test are intentionally limited.

3.2.1 Mathematics Test

For the Mathematics Test, the only external resource that will be provided is a **L^AT_EX cheat sheet**. This is intended for participants who choose to write and format their mathematical solutions using L^AT_EX.

No other external resources may be accessed during the Mathematics Test.

3.2.2 Programming Test

For the Programming Test, participants may access only the official **Python documentation** provided by AOAI.

No other external documentation, tutorials, search engines, forums, or programming resources may be accessed during the test.

Solutions for the Programming Test must be submitted as a `.ipynb` Jupyter Notebook through the Google Form provided for the test.

3.3 Jupyter Notebook Preparation

Participants should be familiar with Jupyter Notebooks before taking the Programming Test. Jupyter Notebooks are widely used in practical AI and machine learning workflows and will be the required format for submitting programming solutions.

The following tutorial is provided as preparation for participants who are not already familiar with Jupyter Notebooks:

Jupyter Notebook Tutorial Dataquest

Participants are expected to be comfortable creating, editing, running, and saving a Jupyter Notebook before the Programming Test.

4 Mathematics Test

The Mathematics Test consists of three problems covering different areas of mathematics relevant to problem solving and Artificial Intelligence.

The problems are designed to assess participants' mathematical reasoning, ability to understand unfamiliar concepts, and ability to apply mathematical ideas to new situations. Some of the concepts introduced in the test may extend beyond the standard Algerian high school curriculum.

Prior knowledge of Artificial Intelligence is not required. The necessary mathematical concepts and definitions will be introduced within the problems themselves.

The three problems will cover:

- **Linear Algebra**
- **Calculus**
- **Probability & Statistics**

The test is intended to evaluate not only computational ability, but also the participant's capacity to read mathematical definitions, derive conclusions from given information, and apply newly introduced concepts to progressively more challenging situations.

5 Programming Test

The Programming Test consists of three problems covering different components commonly encountered in Artificial Intelligence development pipelines.

Python will be the primary programming language used throughout the test. Prior knowledge of Artificial Intelligence is not required. Each problem will provide the necessary context, concepts, specifications, and requirements needed to complete the task.

The three problems will cover different areas of Artificial Intelligence, including:

- **Machine Learning**
- **Computer Vision**
- **Natural Language Processing**

The programming problems are designed to assess the participant's ability to read and understand technical descriptions, extract requirements from a specification, reason about an unfamiliar idea, and translate that understanding into correct and efficient Python code.

Rather than testing memorization of specific Artificial Intelligence libraries or frameworks, the test will emphasize fundamental programming ability, algorithmic thinking, handling of data, and the ability to implement concepts from a given description.

The problems may progressively increase in difficulty and require participants to combine multiple programming concepts. The objective is to stress-test their ability to understand new ideas and implement them correctly under test conditions.

6 What Participants Should Expect

The tasks in AOAI Genesis may introduce concepts that participants have not encountered before.

Participants are therefore not expected to rely solely on previously memorized knowledge. Instead, they should be prepared to read definitions, examples, explanations, and specifications provided in the tasks and use them to solve the problems that follow.

Prior knowledge of Artificial Intelligence is not required for the tests. However, participants should have the mathematical and programming foundations necessary to work with the material.

7 Preparation Material

The resources in this section are provided to define a reasonable and focused scope of preparation for AOAI Genesis.

They are intended to help participants concentrate their preparation on the concepts and level of difficulty relevant to the tests, taking into account the available preparation time and the expected level of high school participants.

These resources are **not the only resources that participants may use**. Students are free to use other books, courses, videos, documentation, or learning materials if they find them more suitable for their level of understanding or preferred way of learning.

Additional miscellaneous resources will also be provided for participants who wish to explore specific topics in greater depth or approach them from a different perspective.

7.1 Mathematics

7.1.1 Linear Algebra

Mathematics for Machine Learning Specialization : Linear Algebra

Imperial College London

Recommended material: The material covering the fundamental concepts of vectors, matrices, linear transformations, and related topics relevant to the Linear Algebra task.

The course playlist can be accessed here:

[**Mathematics for Machine Learning Specialization : Linear Algebra**](#)

7.1.2 Calculus

Mathematics for Machine Learning Specialization : Multivariate Calculus

Imperial College London

Recommended material: The material covering derivatives, partial derivatives, gradients, and optimization concepts relevant to the Calculus task.

The course playlist can be accessed here:

[**Mathematics for Machine Learning Specialization : Multivariate Calculus**](#)

7.1.3 Probability & Statistics

High School Statistics

Khan Academy

Recommended material: The relevant material covering basic probability, conditional probability, independence, tree diagrams, Bayes' theorem, probability distributions, and expected value.

For AOAI Genesis, participants should focus primarily on the following two units:

- **Unit 6: Probability**
- **Unit 7: Probability Distributions and Expected Value**

The complete High School Statistics course can be accessed here:

[**High School Statistics – Khan Academy**](#)

The recommended units can be accessed directly here:

[**Unit 6: Probability**](#)

[**Unit 7: Probability Distributions and Expected Value**](#)

7.1.4 Further Study: Mathematics for Machine Learning Book

Participants who are interested in studying these topics in greater depth, with additional exercises and problems, may also consult the book **Mathematics for Machine Learning (MML)**.

This book provides a more comprehensive treatment of the mathematical foundations used in machine learning and can be used as an additional reference for participants who want to explore the topics beyond the recommended preparation material.

Mathematics for Machine Learning Full Book (PDF)

7.2 Programming

The programming preparation material is intended to provide participants with the Python and general programming foundations required to approach the programming tasks in AOAII Genesis.

The main preparation resource is the **Python track on Exercism**. Participants should first work through the concepts section to become comfortable with the language and its syntax. After completing the relevant concepts, participants are encouraged to solve a selection of exercises to reinforce their understanding through practice.

7.2.1 Python Concepts

The Python concepts section on Exercism covers the fundamental features and syntax of the Python programming language.

Exercism : Python Concepts

Participants are encouraged to go through the concepts in the track before moving on to the exercises below.

7.2.2 Recommended Python Exercises

Reading about the language alone is not sufficient to become comfortable with programming. Participants are therefore encouraged to solve programming exercises after studying the relevant concepts.

The following exercises provide practice with different aspects of Python, including strings, lists, algorithms, matrices, searching, and basic problem-solving:

1. [Isogram](#)
2. [Pangram](#)
3. [Binary Search](#)
4. [List Ops](#)
5. [Pig Latin](#)
6. [Acronym](#)

7. [Series](#)
8. [Matrix](#)
9. [Markdown](#)
10. [Grep](#)
11. [Affine Cipher](#)

Participants do not necessarily need to complete every exercise above. They are provided as a practical set of problems that can help reinforce Python programming skills after studying the language concepts.

7.2.3 Relation to the Programming Test

The Exercism material is intended to build the general Python programming foundation needed for the test. It should **not be taken as a representation of the style or content of the actual competition tasks**.

The programming tasks in AOAI Genesis will be directly related to concepts and operations that arise in Artificial Intelligence pipelines. The concepts implemented in the test will therefore be different from the general-purpose programming exercises found on Exercism.

Participants should focus on becoming comfortable with Python itself, including understanding its syntax, data structures, functions, and basic algorithmic problem-solving. The ability to apply these foundations to AI-related problems will then be assessed during the test.

7.2.4 Further Study: Introduction to Machine Learning

Participants who are willing to put in additional effort and begin studying Artificial Intelligence and implementing AI-related concepts from scratch may also explore the course materials from MIT's **Introduction to Machine Learning** course.

The course provides additional lectures, homework, and labs exercises that can help participants move beyond general Python programming and begin working directly with machine-learning concepts.

MITx : Introduction to Machine Learning

This material is **optional** and is not required for AOAI Genesis. It is provided for participants who want to go beyond the minimum preparation scope and begin developing a deeper understanding of Artificial Intelligence algorithms.

8 Additional Information

Further information concerning AOAI Genesis will be announced as it becomes available.

If any part of the information provided in this document is unclear, or if participants have questions regarding the tests, preparation material, or other aspects of AOAI Genesis, they are encouraged to contact me directly for clarification.

Participants may also ask questions in the general AOAI Discord server, where other participants can share their understanding and help each other with general questions.